ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

CONCEPT RECAPITULATION TEST - II

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. There are some passengers inside a stationary railway compartment. The centre of mass of the compartment itself (without the passengers) is C_1, while the centre of mass of the 'compartment plus passengers' system is C_2. If the passengers move about inside the compartment,
(A) both C_1 and C_2 will move with respect to the ground
(B) neither C_1 nor C_2 will move with respect to the ground
(C) C_1 will move but C_2 will stationary with respect to the ground
(D) C_2 will move but C_1 will be stationary with respect to the ground

2. A false balance has equal arms. An object weighs x when placed in one pan and y in the other pan. The true weight of the object is equal to
(A) $\sqrt{xy}$ (B) $\frac{x+y}{2}$
(C) $\frac{x^2+y^2}{2}$ (D) $\frac{\sqrt{x^2+y^2}}{2}$

3. Two identical cylindrical vessels with their bases at the same level, each contains a liquid of density ρ. The height of the liquid in one vessel is h_1 and that in the other h_2. The area of either base is A. The work done by gravity in equalizing the levels when the vessels are inter connected is
(A) $A\rho g\left(\frac{h_1-h_2}{2}\right)$ (B) $A\rho g\left(\frac{h_1-h_2}{2}\right)^2$
(C) $A\rho g\left(\frac{h_1-h_2}{4}\right)$ (D) $A\rho g\left(\frac{h_1-h_2}{4}\right)^2$

4. Dimensional formula for Plank's constant is identical to that of
(A) Torque (B) Power
(C) Linear momentum (D) Angular momentum

Space for Rough work

5. A wheel of radius r rolls without slipping with a speed v on a horizontal road. When it is at a point A on the road, a small blob of mud separates from the wheel at its highest point and lands at point B on the road

(A) $AB = v\sqrt{\frac{r}{g}}$ (B) $AB = 2v\sqrt{\frac{r}{g}}$

(C) $AB = 4v\sqrt{\frac{r}{g}}$ (D) $AB = 8v\sqrt{\frac{r}{g}}$

6. A particle moves along the parabolic path $y = ax^2$ in such a way that the x-component of the velocity remains constant, say c. The acceleration of the particle is

(A) $ac\hat{k}$ (B) $2ac^2\hat{j}$

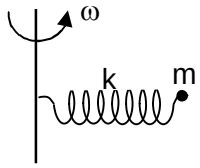
(C) $ac^2\hat{k}$ (D) $a^2c^2\hat{j}$

7. A radioactive nucleus of mass number A, initially at rest, emits an α particle with speed v. The recoil speed of the daughter nucleus is

(A) $\frac{4v}{A-4}$ (B) $\frac{4v}{A}$

(C) $\frac{(A-4)v}{A}$ (D) $\frac{(A-4)v}{4}$

8. A particle of mass m is fixed to one end of a light spring of force constant k and unstretched length l. The system is rotated about the other end of the spring with an angular velocity ω, in gravity free space. The increase in length of the spring will be



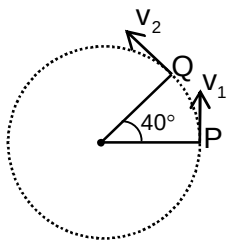
(A) $\frac{ml\omega^2}{k}$ (B) $\frac{ml\omega^2}{k-m\omega^2}$

(C) $\frac{ml\omega^2}{k+m\omega^2}$ (D) None of these

9. A particle is moving in a circle of radius r centred at O with constant speed v. The change in velocity in moving from P to Q ($\angle POQ = 40°$) is

(A) 2vcos40° (B) 2vsin40°

(C) 2vcos20° (D) 2vsin20°



Space for Rough work

10. The electric potential (in volt) in a region is given by
$V = 6x - 8xy^2 - 8y + 6yz - 4x^2$
Then electric force acting on a point charge of 2C placed at the origin will be
(A) 2 N (B) 6 N
(C) 8 N (D) 20 N

11. Two containers of equal volume contain identical gases at pressures P_1 and P_2 and absolute temperatures T_1 and T_2 respectively. The vessels are joined and the gas reaches a common pressure P and a common temperature T. Then
(A) $P = \left(\frac{P_1}{T_1} + \frac{P_2}{T_2}\right)T$ (B) $P = \frac{1}{2}\left(\frac{P_1}{T_1} + \frac{P_2}{T_2}\right)T$
(C) $P = \left(\frac{P_1T_2 + P_2T_1}{T_1^2 + T_2^2}\right)T$ (D) $P = \left(\frac{P_1T_2 + P_2T_1}{T_1^2 - T_2^2}\right)T$

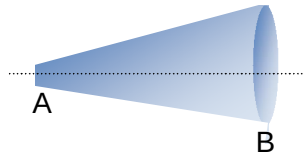
12. The half life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially both of them have same number of atoms. Then,
(A) X and Y have the same decay rate initially.
(B) X and Y decay at the same rate always.
(C) Y will decay at a faster rate than X.
(D) X will decay at a faster rate than Y.

13. The temperature of source and sink of a Carnot Engine are 327°C and 27°C respectively. The efficiency of the engine is
(A) $1-\left(\frac{27}{327}\right)$ (B) $\frac{27}{327}$
(C) 0.5 (D) 0.7

14. For a gas $\frac{R}{C_P} = \frac{2}{3}$. Possibly the gas is
(A) diatomic (B) monatomic
(C) a mixture of diatomic and monatomic (D) polyatomic

Space for Rough work

15. A wire has a non-uniform cross-section as shown in the figure. A steady current is flowing through it. Then the drift speed of the electrons
(A) is constant throughout the wire
(B) decreases from A to B
(C) increases from A to B
(D) varies randomly



16. Which of the following is most suitable for the core of the electromagnets?
(A) Air
(B) Steel
(C) Soft iron
(D) Cu-Ni alloy

17. A black body radiates power P and maximum energy is radiated by it around a wavelength λ_0. The temperature of the black body is now changed such that it radiates maximum energy around the wavelength $\frac{3\lambda_0}{4}$. The power radiated by it now is
(A) $\frac{256}{81}P$
(B) $\frac{16}{9}P$
(C) $\frac{64}{27}P$
(D) $\frac{4}{3}P$

18. The plane of the dip circle is set in the geographical meridian and apparent dip is θ_1. It is then set in a vertical plane perpendicular to the geographical meridian, the apparent dip becomes θ_2. The angle of declination α at that place is given by
(A) $\tan\alpha = \sqrt{\tan\theta_1 \tan\theta_2}$
(B) $\tan\alpha = \sqrt{\tan^2\theta_1 + \tan^2\theta_2}$
(C) $\tan\alpha = \frac{\tan\theta_1}{\tan\theta_2}$
(D) $\tan\alpha = \frac{\tan\theta_2}{\tan\theta_1}$

19. According to Faraday's law of electromagnetic induction
(A) the direction of induced current is such that it opposes the cause producing it.
(B) the magnitude of induced emf produced in a coil is directly proportional to the rate of change of magnetic flux.
(C) The direction of induced emf is such that it opposes the cause producing it.
(D) None of the above.

Space for Rough work

20. If ϕ is phase difference between current and voltage, the wattless component of current is
(A) I cosϕ (B) I sinϕ
(C) I tanϕ (D) I cos$^2\phi$

21. A particle is vibrating in S.H.M. Its velocities are v_1 and v_2 when the displacements from the mean position are y_1 and y_2, respectively, then its time period is
(A) $2\pi\sqrt{\frac{y_1^2+y_2^2}{v_1^2+v_2^2}}$ (B) $2\pi\sqrt{\frac{v_1^2+v_2^2}{y_1^2+y_2^2}}$
(C) $2\pi\sqrt{\frac{v_2^2-v_1^2}{y_1^2-y_2^2}}$ (D) $2\pi\sqrt{\frac{y_1^2-y_2^2}{v_2^2-v_1^2}}$

22. The first overtone of an open pipe has frequency n. The first overtone of a closed pipe of the same length will have frequency
(A) $\frac{n}{2}$ (B) 2n
(C) $\frac{3n}{4}$ (D) $\frac{4n}{3}$

23. **Statement-1:** The resolving power of a telescope is more if the diameter of the objective lens is more.
Statement-2: Objective lens of large diameter collects more light.
(A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.
(C) Statement-1 is true; Statement-1 is false.
(D) Statement-1 is false; Statement-1 is true.

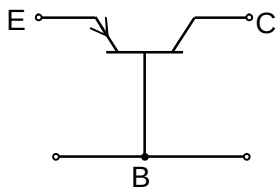
24. Light of wavelength 6328 Å is incident normally on a slit having a width of 0.2 mm. the distance of the screen from the slit is 0.9 m. The angular width of the central maximum is
(A) 0.09 degree (B) 0.72 degree
(C) 0.18 degree (D) 0.36 degree

25. A clear sheet of polaroid is placed on the top of similar sheet so that their axes make an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with each other. The ratio of intensity of the emergent light to that of unpolarised incident light is
(A) 16 : 25 (B) 9 : 25
(C) 4 : 5 (D) 8 : 25

Space for Rough work

26. The vernier constant of a travelling microscope is 0.001 cm. If 49 main scale divisions coincide with 50 main scale divisions, then the value of 1 main scale division is
(A) 0.1 mm (B) 0.5 mm
(C) 0.4 mm (D) 1 mm

27. For circuit shown in figure I_E = 4 mA, I_B = 40μA. What are the values of α and I_C?
(A) 0.99, 3.96 mA
(B) 1.01, 4.04 mA
(C) 0.97, 4.04 mA
(D) 0.99, 4.04 Ma



28. The average value of electric energy density in an electromagnetic wave is (E_0 is peak value)
(A) $\frac{1}{2}\varepsilon_0 E_0^2$ (B) $\frac{E_0^2}{2\varepsilon_0}$
(C) $\varepsilon_0 E_0^2$ (D) $\frac{1}{4}\varepsilon_0 E_0^2$

29. A body is projected away from the earth with a speed 3v, where v, is the escape velocity. The speed of the body at infinity will be
(A) v_e (B) $\sqrt{2}v_e$
(C) $2v_e$ (D) $2\sqrt{2}v_e$

30. A vessel of cross-sectional area A contains a liquid to a height H_1. If a hole having cross-sectional area a is made at the bottom of the vessel, then the time taken by the liquid level to decrease from H_1 and H_2 is
(A) $\frac{A}{a}\sqrt{\frac{g}{2}}\left[\sqrt{H_1}-\sqrt{H_2}\right]$ (B) $\frac{A}{a}\sqrt{\frac{2}{g}}\left[\sqrt{H_1}-\sqrt{H_2}\right]$
(C) $\frac{a}{A}\sqrt{\frac{g}{2}}\left[\sqrt{H_1}-\sqrt{H_2}\right]$ (D) $\frac{a}{A}\sqrt{\frac{2}{g}}\left[\sqrt{H_1}-\sqrt{H_2}\right]$

Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Which of the following tests you would perform to identify the functional group present in salicylic acid?
   (A) $FeCl_3$ and $NaHCO_3$ test (B) $FeCl_3$ and NaOH test
   (C) $FeCl_3$ and 2, 4-dinitrophenylhydrazine test (D) $FeCl_3$ and Schiff's reagent test

2. A metal oxide is yellow when hot and white when cold. The metal oxide is
   (A) CuO (B) ZnO
   (C) PbO (D) All

3. A metal chloride solution on mixing with K_2CrO_4 solution gives a yellow ppt. insoluble in acetic acid. The metal may be.
   (A) Mercury (B) Zinc
   (C) Silver (D) Lead

4. Enthalpy of dissociation of water in kJ mol^{-1} is
   (A) +13.7 (B) +57.3
   (C) +18.0 (D) $+18 \times 4.2$

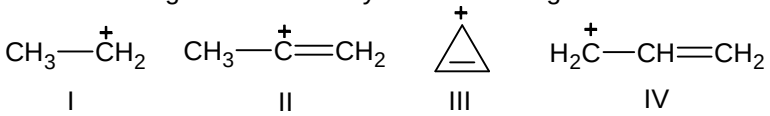
5. For the preparation of iodoform from acetone; we require
   (A) KI (B) KI_3
   (C) KOI (D) KIO_3

6. Cl_2 gas passed through Lassaigne's extract containing CCl_4. If the extract contains both NaBr and NaI then which colour will appear in the CCl_4 layer?
   (A) Violet colour (B) Brown colour
   (C) Green colour (D) Yellow colour

Space for Rough work

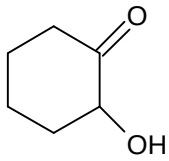
7. Which one of the following does not involve peptisation?
(A) $Fe(OH)_3$ precipitate shaken with a small amount of dil. HCl
(B) $Fe(OH)_3$ precipitate shaken with $FeCl_3$ solution
(C) $AuCl_3$ solution shaken with $SnCl_2$ solution
(D) $AgNO_3$ solution shaken with KI solution

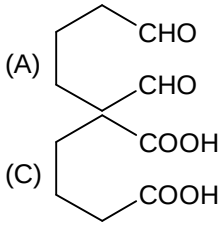
8. The increasing order of stability of the following carbocation is



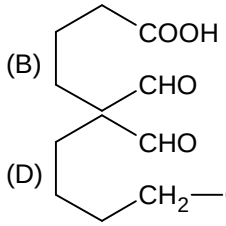
I II III IV

(A) IV < I < III < II (B) II < I < IV < III
(C) I < II < III < IV (D) III < I < IV < II

9. The product obtained when  is oxidized with HIO_4 is



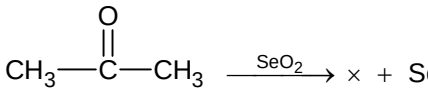
(A)

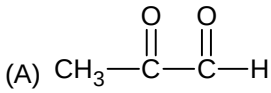
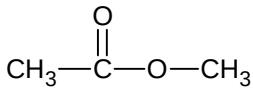
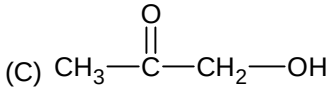


(B)

(C)

(D)

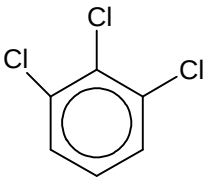
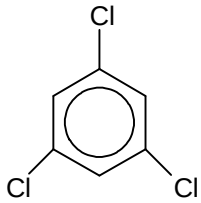
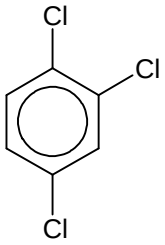
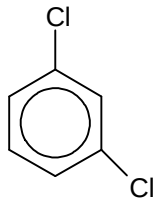
10.  here, X is:

(A) 
(B) 
(C) 
(D) None of the above

11. The weight of a molecule of the compound $C_{60}H_{122}$ is
(A) 1.4×10^{-21} g (B) 1.09×10^{-21} g
(C) 5.025×10^{23} g (D) 16.023×10^{23} g

Space for Rough work

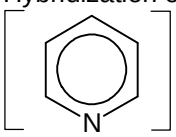
12. Benzoic acid undergoes dimerisation in benzene solution. The van't Hoff factor (i) is
(A) $i = (1 - x)$ (B) $i = (1 + x)$
(C) $i = (1 - x/2)$ (D) $i = (1 + x/2)$

13. Which of the following has maximum dipole moment?
(A) 
(B) 
(C) 
(D) 

14. The ONO bond angle is maximum in
(A) NO_3^- (B) NO_2^-
(C) NO_2 (D) NO_2^+

15. Calculate the partial pressure of carbon monoxide from the following data
$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g); \quad K_p = 8 \times 10^{-2}$
$CO_2(g) + C(s) \longrightarrow 2CO(g); \quad K_p = 2$
(A) 0.2 (B) 0.4
(C) 1.6 (D) 4

16. The diagonal partner of element B is
(A) Li (B) Al
(C) Si (D) Mg

17. Hybridization of the nitrogen atom and electron geometry around nitrogen atom in pyridine
 is
(A) Sp^2, planar triangular (B) Sp^3, pyramidal
(C) Sp^3, tetrahedral (D) Sp^2, v-shape

Space for Rough work

18. The energy of the second Bohr orbit of the hydrogen atom is –328 kJ mol^{-1}; hence the energy of fourth Bohr orbit would be
(A) –1312 kJ mol^{-1} (B) –82 kJ mol^{-1}
(C) –41 kJ mol^{-1} (D) –164 kJ mol^{-1}

19. The ionic radii of isoelectronic species N^{3-}, O^{2-} and F^{-} in Å are in the order
(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40
(C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40

20. What will be the correct order of vapour pressure of water, acetone and ether at 30ºC. Given that among these compounds, water has maximum boiling point and ether has minimum boiling point?
(A) Water < ether < acetone (B) Water < acetone < ether
(C) Ether < acetone < water (D) Acetone < ether < water

21. Conjugate base of H_2 is
(A) H_3^+ (B) H_3^-
(C) H^+ (D) H^-

22. How many ml of 0.05 M $KMnO_4$ (acid) are required to oxidize 2.0 gram of $FeSO_4$ in dilute solution?
(A) 52.63 (B) 42.63
(C) 12.63 (D) 21.36

23. If the bond lengths of C ≡ C, C ≡ N are x and y Å respectively N ≡ N bond length is given by
(A) x + 2y (B) $\frac{x-2y}{2}$
(C) x + y (D) 2y – x

24. Which of the following complex species does not involve inner orbital hybridization?
(A) $[CoF_6]^{3-}$ (B) $[Co(NH_3)_6]^{3+}$
(C) $[Fe(CN)_6]^{3-}$ (D) $[Cr(NH_3)_6]^{3+}$

Space for Rough work

25. For the reaction,

$A(g) + 2B(g) \longrightarrow 2C(g) + 3D(g)$

The value of ΔH at 27ºC is 19.0 K cal. The value of ΔE for the reaction would be (given R = 2.0 cal K^{-1} mol^{-1})

(A) 20.8 Kcal (B) 19.8 Kcal
(C) 18.8 Kcal (D) 17.8 Kcal

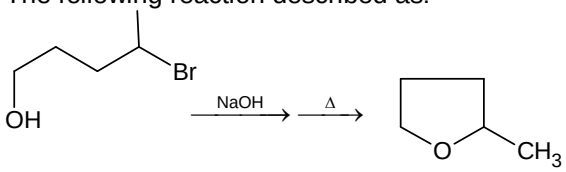
26. For the reaction

$C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

The k_p = 0.05 atm. The value of ΔG^o of the reaction at 627ºC would be

(A) 11.19 kJ mol^{-1} (B) 22.40 kJ mol^{-1}
(C) 33.57 kJ mol^{-1} (D) 27.98 kJ mol^{-1}

27. The following reaction described as:



(A) S_N1 reaction with racemisation
(B) Intramolecular S_N2 reaction with Walden inversion
(C) Intramolecular S_N2 reaction with retention of configuration
(D) Intramolecular S_N1 reaction with racemisation

28. 75% of the first order reaction was completed in 32 min. 50% of the reaction was completed in

(A) 24 min (B) 8 min
(C) 16 min (D) 4 min

29. pH of a solution produced when an aqueous solution of pH 6 is mixed with an equal volume of an aqueous solution of pH 3 is about

(A) 4.5 (B) 4.0
(C) 4.3 (D) 3.3

30. $2\,CH_3-\overset{O}{\overset{\|}{C}}OC_2H_5 \xrightarrow{C_2H_5ONa} A$, which is true about A.

(A) A forms oxime (B) A shows tautomerism
(C) A shows idoform test (D) All of them

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The value of $\int_{-1}^{1} \tan^{-1}\left[x^2+\frac{1}{2}\right]dx+\int_{-1}^{1} \cot^{-1}\left[x^2-\frac{1}{2}\right]dx$ is equal to (where [.] denotes greatest integer function)

(A) $\frac{3\pi}{4}\left(1-\frac{1}{\sqrt{2}}\right)$ (B) $\frac{3\pi}{4}\left(1+\frac{1}{\sqrt{2}}\right)$

(C) $\frac{\pi}{4}\left(1-\frac{1}{\sqrt{2}}\right)$ (D) none of these

2. The number of ordered pairs of positive integers (m, n) satisfying $m \le 2n \le 60$, $n \le 2m \le 60$ is

(A) 240 (B) 480

(C) 900 (D) none of these

3. Let 'P' be a point which does not lie outside the triangle ABC, A ≡ (3, 2), B ≡ (0, 0), C ≡ (0, 4) and satisfies d(P, A) ≥ maximum {d(P, B), d (P, C)}, then maximum distance of P from side BC where d(P, A) gives the distance between P and A, is

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$

(C) 3 (D) 0

4. The mth term of an arithmetic progression is x and the nth term is y. Then the sum of the first (m + n) terms is

(A) $\frac{m+n}{2}\left[(x+y)+\frac{x-y}{m-n}\right]$ (B) $\frac{m+n}{2}\left[(x-y)+\frac{x+y}{m-n}\right]$

(C) $\frac{1}{2}\left[\frac{x+y}{m+n}+\frac{x-y}{m-n}\right]$ (D) $\frac{1}{2}\left[\frac{x+y}{m+n}-\frac{x-y}{m-n}\right]$

Space for rough work

5. Let the $f(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is

(A) Continuous but not differentiable at x = 1 (B) Differentiable at x = 1
(C) Neither continuous nor differentiable at x = 1 (D) Continuous everywhere

6. $\lim_{n\to\infty} \frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}}$ is equal to

(A) 5 (B) 3
(C) 1 (D) zero

7. If the point $P(x_1+t(x_2-x_1), y_1+t(y_2-y_1))$ divides the join of $A(x_1,y_1)$ and $B(x_2,y_2)$ internally then

(A) t < 0 (B) 0 < t < 1
(C) t > 1 (D) t = 1

8. If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in G. P with the same common ratio, then the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3)

(A) Lie on a straight line (B) Lie on an ellipse
(C) Lie on a circle (D) are vertices of a triangle

9. If in a ΔABC, cos A + 2 cos B + cos C = 2, then a, b, c are in

(A) AP (B) HP
(C) GP (D) none of these

10. If $f(x) = \int_0^1 \frac{x^2+t^2}{2-t}\,dt$, then the curve y = f(x) represents a

(A) Straight line (B) Parabola
(C) Hyperbola (D) None of these

11. Let P and Q be points (4, –4) and (9, 6) of the parabola $y^2 = 4a(x - b)$. Let R be a point on the arc of the parabola between P and Q. Then the area of ΔPRQ is largest when

(A) $\angle PRQ = 90°$ (B) The point R is (4, 4)
(C) The point R is $\left(\frac{1}{4}, 1\right)$ (D) none of these

Space for rough work

12. For what values of a, m and b, Lagrange's mean value theorem is applicable to the function f(x) for $x \in [0, 2]$, $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + a & 0 < x < 1 \\ mx + b & 1 \le x \le 2 \end{cases}$

(A) a = 3, m = – 2, b = 0
(B) a = 3, m = – 2, b = 4
(C) a = 3, m = 2, b = 0
(D) No such a, m, b exist

13. Eccentricity of the hyperbola whose asymptotes are given by 3x + 2y – 5 = 0 and 2x – 3y – 5 = 0 is

(A) $\sqrt{2}$
(B) $\frac{3}{2}$
(C) 2
(D) $\frac{5}{3}$

14. Consider the ellipse $\frac{x^2}{3} + \frac{y^2}{1} = 1$. Let P, Q, R, S be four points on this ellipse such that the normals drawn from these points are concurrent at (2, 2) then the centre of the conic (apart from the given ellipse) on which these 4 points lie is
(A) P, Q, R, S lie only on the given ellipse and on no other conic
(B) (3, –1)
(C) (3, 1)
(D) (0, 0)

15. Let $f(x) = \int_0^x \cos\left(\frac{t^2 + 2t + 1}{5}\right) dt$, $0 < x < 2$. Then f(x)

(A) increases monotonically
(B) decreasing monotonically
(C) has one point of local maximum
(D) has one point of local minimum

16. If $I = \int \frac{x^5 dx}{\sqrt{1 + x^3}}$, then I is equal to

(A) $\frac{2}{9}(1 + x^3)^{\frac{5}{2}} + \frac{2}{3}(1 + x^3)^{\frac{3}{2}} + c$
(B) $\frac{2}{9}(1 + x^3)^{\frac{3}{2}} - \frac{2}{3}(1 + x^3)^{\frac{1}{2}} + c$
(C) $\log\left|\sqrt{x} + \sqrt{1 + x^3}\right| + c$
(D) $x^2 \log(1 + x^3) + c$

Space for rough work

17. If w is an imaginary cube root of unity, then value of the expression $2(1 + w)(1 + w^2) + 3(2 + w)(2 + w^2) + ... + (n + 1)(n + w)(n + w^2)$ is

(A) $\frac{1}{4}n^2(n+1)^2 + n$ (B) $\frac{1}{4}n^2(n+1)^2 - n$

(C) $\frac{1}{4}(n+1)^2 - n$ (D) none of these

18 If z be a complex number satisfying $|z|^2 + 2(z + \bar{z}) - 5 = 0$, then complex number z + 3 + 2i will lie on

(A) Circle with centre 1 – 2i radius 4 (B) Circle with centre 1 + 2i radius 4

(C) Circle with centre 1 + 2i radius 3 (D) Circle with centre 1 – 2i radius 3

19. If one of the roots of the equation $\begin{vmatrix} 7 & 6 & x^2-13 \\ 2 & x^2-13 & 2 \\ x^2-13 & 3 & 7 \end{vmatrix} = 0$ is x = 2, then sum of all other five roots is

(A) $2\sqrt{15}$ (B) –2

(C) $\sqrt{20} + \sqrt{15} - 2$ (D) none of these

20. f: R $\to$ R, f(x) is differentiable such that $f(f(x)) = k(x^5 + x)$, $(k \neq 0)$, then f(x) is always

(A) Increasing (B) Decreasing

(C) Either increasing or decreasing (D) Non-monotonic

21. Let $\vec{a}$ and $\vec{b}$ are two perpendicular unit vectors. If $\vec{c}$ is another unit vector equally inclined at angle θ to the vectors $\vec{a}$ and $\vec{b}$, then the set of exhaustive values of θ in $[0, 2\pi]$ is

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left[0, \frac{\pi}{4}\right]$

(C) $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$ (D) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

22. Let A, B, C be point with position vectors $r_1 = 2\hat{i} - \hat{j} + \hat{k}$, $r_2 = \hat{i} + 2\hat{j} + 3\hat{k}$ and $r_3 = 3\hat{i} + \hat{j} + 2\hat{k}$ relative to the origin 'O'. The shortest distance between point B and plane OAC is

(A) 10 (B) 5

(C) $\sqrt{\frac{5}{7}}$ (D) $2\sqrt{\frac{5}{7}}$

Space for rough work

23. The maximum number of regions in which 10 circles can divide a plane is
(A) 512 (B) 1024
(C) 2048 (D) None of these

24. If [x] and {x} denotes the greatest integer and fractional part function, then the number of real x, satisfying the equation $(x - 2)[x] = \{x\} - 1$, is
(A) 0 (B) 1
(C) 2 (D) infinite

25. Let g(x) = ax + b, where a < 0 and g is defined from [1, 3] to [0, 2] then the value of $\cot\left(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|)\right)$ is equal to
(A) g(1) (B) g(2)
(C) g(3) (D) g(1) + g(3)

26. Consider the 12 face diagonals of a rectangular block. How many pairs of them are skew lines?
(A) 24 (B) 30
(C) 48 (D) 60

27. Suppose $M = \{(x, y) | |xy| = 1, x > 0\}$ and $N = \{(x, y) | \tan^{-1} x + \tan^{-1} y = \pi\}$ which of the following statements is true?
(A) $M \cup N = \{(x, y) | |xy| = 1\}$
(B) $M \cup N = M$
(C) $M \cup N = N$
(D) $M \cup N = \{(x, y) | |xy| = 1, x, y$ are non negative simultaneously$\}$

28. Let F(x) be such that $F\left(\frac{1-x}{1+x}\right) = x$ for all $x \neq -1$ which of the following statements is true?
(A) $F(-2 - x) = -2 - F(x)$ (B) $F(-x) = F\left(\frac{1-x}{1+x}\right), x \neq 1$
(C) $F\left(\frac{1}{x}\right) = F(x), x \neq 0$ (D) $F(F(x)) = -x$

29. Which of the following is logically equivalent to $p \rightarrow (p \vee q)$?
(A) $\sim p \wedge (p \vee q)$ (B) $q \rightarrow (p \rightarrow q)$
(C) $\sim p \wedge \sim q$ (D) none of these

30. The mean of the 200 observation is 48 and their standard deviation is 3. Then sum of the squares of all the observations
(A) 46200 (B) 42600
(C) 62400 (D) none of these

Space for rough work